

BALTIMORE DOWNTOWN, MD, USA

WMO: 745944

Lat: 39.281N

Lon: 76.611W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93784

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.5	(c) -6.3	(d) -20.2	(e) 0.6	(f) -7.0	(g) -17.7	(h) 0.8	(i) -5.0	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	35.3	24.2	33.8	23.5	32.5	23.0	25.6	32.4	25.0	31.1	24.4	30.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	18.7	28.4	23.2	17.9	27.9	22.6	17.3	27.7	78.6	32.5	75.6	31.0	73.2	30.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -10.6	(f) 37.5	(g) 2.6	(h) 2.1	(i) -12.5	(j) 39.0	(k) -14.1	(l) 40.2	(m) -15.6	(n) 41.4	(o) -17.5	(p) 42.9		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-12.3	26.9	2.4	0.8	-14.0	27.5	-15.4	28.0	-16.8	28.5	-18.5	29.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	2.5	3.7	8.2	14.5	19.6	24.9	27.3	26.4	22.6	16.2	10.2	5.2
	DBStd	9.71	5.50	5.02	5.11	4.79	4.40	3.41	2.85	2.64	3.49	4.49	4.42	4.63
	HDD10.0	747	241	183	96	12	0	0	0	0	0	4	50	161
	HDD18.3	2143	492	408	314	135	37	2	0	0	6	97	244	408
	CDD10.0	2637	7	8	42	147	298	447	538	507	379	195	57	12
	CDD18.3	992	0	0	2	19	78	199	279	249	135	29	1	0
	CDH23.3	9394	0	2	16	173	622	1912	3108	2433	958	166	3	1
	CDH26.7	3691	0	0	2	54	209	755	1407	939	286	38	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	1067	78	79	98	79	100	78	96	105	92	75	88	94
	PrecMax	1564	220	176	248	192	236	254	297	377	223	220	222	205
	PrecMin	751	10	17	24	9	15	16	18	8	2	3	11	18
	PrecStd	223	44	42	54	42	52	45	54	76	61	48	49	56
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	20.8	25.5	31.5	33.8	36.4	38.0	36.1	33.8	30.1	23.5	20.1
		MCWB	14.0	14.0	16.2	18.8	22.7	24.1	25.2	24.7	23.7	21.4	17.0	16.2
	2%	DB	15.4	17.0	21.6	28.0	31.1	34.2	35.8	34.2	31.8	27.3	20.8	16.6
		MCWB	12.0	10.9	14.1	18.0	22.0	23.3	24.5	24.1	22.5	20.5	15.2	13.6
	5%	DB	12.5	13.4	18.6	24.9	29.0	32.4	34.2	32.6	29.8	24.9	18.5	14.0
		MCWB	8.9	8.5	12.3	16.4	20.5	22.5	23.7	23.4	21.7	19.4	13.8	11.0
	10%	DB	9.9	10.7	16.1	22.1	26.9	30.7	32.7	31.2	28.1	22.9	16.7	11.7
		MCWB	6.3	6.2	10.6	14.9	19.4	21.9	23.3	22.8	21.0	18.0	12.6	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.6	17.4	20.7	24.1	25.4	26.7	26.5	25.5	23.8	18.5	17.1
		MCDB	18.2	19.1	23.6	27.3	31.8	33.0	35.0	33.4	30.6	26.5	21.8	18.6
	2%	WB	12.6	12.1	15.5	19.1	22.9	24.5	25.8	25.6	24.5	22.0	16.7	14.1
		MCDB	15.0	15.3	20.3	25.4	29.5	31.3	33.1	31.6	29.0	26.0	19.5	16.2
	5%	WB	9.4	9.0	13.5	17.7	21.8	23.9	25.1	24.8	23.6	20.6	15.0	11.4
		MCDB	12.0	12.8	17.6	23.6	27.6	30.2	31.7	30.2	27.7	24.3	17.6	13.4
	10%	WB	6.9	6.7	11.3	16.2	20.6	23.2	24.5	24.1	22.6	18.6	13.4	8.9
		MCDB	9.4	10.1	15.2	21.4	25.6	28.8	30.5	28.9	26.6	22.1	16.1	11.3
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.4	8.3	9.3	8.6	8.6	8.3	7.8	7.7	7.8	7.4	6.4
		MCDBR	10.6	11.5	12.1	13.1	11.0	10.4	9.9	9.5	9.6	9.9	9.8	9.2
		MCWBR	8.3	8.1	7.6	6.2	5.3	4.1	3.5	3.6	4.1	5.0	6.5	7.9
	5% WB	MCDBR	10.0	10.9	11.6	11.5	10.1	9.1	8.8	8.3	7.8	8.0	8.4	8.5
		MCWBR	8.7	8.8	8.3	6.3	5.1	4.0	3.5	3.6	4.2	5.1	6.8	8.2
Clear-Sky Solar Irradiance	taub	0.312	0.322	0.362	0.401	0.446	0.501	0.519	0.514	0.430	0.365	0.326	0.311	
	taud	2.465	2.419	2.374	2.285	2.193	2.083	2.049	2.046	2.253	2.441	2.506	2.505	
	Ebn at Noon	861	900	893	875	839	791	772	763	811	840	837	835	
	Edh at Noon	80	95	111	129	145	162	166	162	124	93	76	72	
All-Sky Solar Radiation	RadAvg	2.02	2.88	3.81	4.90	5.48	6.08	6.02	5.33	4.35	3.21	2.31	1.77	
	RadStd	0.17	0.33	0.28	0.40	0.66	0.47	0.30	0.30	0.51	0.35	0.20	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (79 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+73		

Nomenclature: See separate page